

Experience

Osmo

Sep 2022 - Present

VP, Engineering & Research

Jan 2026 - Present

Lead engineering and research across AI, product engineering, science, and data operations; our systems power a robotic fragrance factory and its customer experience.

Director, Applied Research

Jan 2025 - Jan 2026

Led ML researchers and human-data experts to land Osmo's first multimillion-dollar enterprise program.

Manager, Machine Learning

Jan 2024 - Jan 2025

Invented scent prompting, taught an LLM to write fragrance formulas, and reconstructed the scent of a plum from GC-MS data with analytical chemists.

Founding Research Engineer

Sep 2022 - Jan 2024

Rebuilt the ML stack in PyTorch, PyG, W&B, and GCP, and developed a GNN for molecular potency prediction.

Google

May 2018 - Aug 2022

Student Researcher, Olfaction

Jun 2020 - Aug 2022

Developed machine learning for olfaction that helped seed Osmo, leading to four publications and a utility patent.

Student Researcher, Genomics

May 2019 - May 2020

Built a parallel structural-variant method about 100x faster than the prior state of the art.

Research Intern, Cellular Imaging

May 2018 - Aug 2018

Built a GAN to remove batch effects in high-content cellular imaging; published in Bioinformatics and released through TF-GAN.

DeepMind

Sep 2021 - Dec 2021

Intern, AlphaFold

2021

Trained protein annotation models in JAX using structural representations learned by AlphaFold2.

Uber

May 2016 - Aug 2017

Software Engineer Intern, Machine Learning

Summer 2017

Built a sequence model from mobile sensor data to infer Uber Eats delivery events; won first place at Uber's internal ML poster session.

Software Engineering Intern, Mobile

Summer 2016

Built a web forensics tool for mobile UI testing and an automated failure-report workflow.

Education

University of Illinois Urbana Champaign

2017 - 2022

Doctor of Philosophy in Computer Science

Dissertation: Machine learning for drug discovery and beyond (advisor: *Jian Peng*)

GPA: 4.00 / 4.00 | University Fellowship | Richard T. Cheng Endowed Fellowship

Brandeis University

2013 - 2017

Bachelor of Science in Computer Science and Neuroscience

GPA: 3.96 / 4.00 | Summa Cum Laude | Phi Beta Kappa (Junior) | Schiff Fellowship

Publications (*equal contribution)

- **Foundation models for discovery and exploration in chemical space**

arXiv (2025)

Alexius Wadell, Anoushka Bhutani, Victor Azumah, Austin R. Ellis-Mohr, Andrew J. Stier, Kareem Hegazy, Alexander Brace, Hancheng Zhao, Celia Kelly, Anuj K. Nayak, Yuhang Chen, Dimitrios Simatos, Hongyi Lin, Murali Emani, Venkatram Vishwanath, Kevin Gering, Melisa Alkan, Tom Gibbs, Jack Wells, **Wesley W. Qian**, Richard C. Gerkin, Benjamin Amorelli, Alexander B. Wiltchko, Lav R. Varshney, Bharath Ramsundar, Karthik Duraisamy, Michael W. Mahoney, Arvind Ramanathan, Venkatasubramanian Viswanathan

- **A deep learning and digital archaeology approach for mosquito repellent discovery**

Chemical Senses (2025)

Jennifer N. Wei*, Carlos Ruiz*, Marnix Vlot*, Benjamin Sanchez-Lengeling, Brian K. Lee, Luuk Berning, Martijn W. Vos, Rob WM Henderson, **Wesley W. Qian**, Jacob N. Sanders, D. Michael Ando, Kurt M. Groetsch, Richard C. Gerkin, Alexander B. Wiltchko, Jeffrey A. Riffell, Koen J. Dechering

- **New York Smells: a large multimodal dataset for olfaction**

arXiv (2025)

Ege Ozguroglu, Junbang Liang, Ruoshi Liu, Mia Chiquier, Michael DeTienne, **Wesley W. Qian**, Alexandra Horowitz, Andrew Owens, Carl Vondrick

- **Pervasive mislocalization of pathogenic coding variants underlying human disorders**

Cell (2024)

Jessica Lacoste, Marzieh Haghighi, Shahan Haider, Chloe Reno, Zhen-Yuan Lin, Dmitri Segal, **Wesley W. Qian**, Xueting Xiong, Tanisha Teelucksingh, Esteban Miglietta, Hamdah Shafqat-Abbasi, Pearl V. Ryder, Rebecca Senft, Beth A. Cimini, Ryan R. Murray, Chantal Nyirakanani, Tong Hao, Gregory G. McClain, Frederick P. Roth, Michael A. Calderwood, David E. Hill, Marc Vidal, S Stephen Yi, Nidhi Sahni, Jian Peng, Anne-Claude Gingras, Shantanu Singh, Anne E. Carpenter, Mikko Taipale

- **A principal odor map unifies diverse tasks in human olfactory perception**

Science (2023)

Brian K. Lee*, Emily E Mayhew*, Benjamin Sanchez-Lengeling, Jennifer N. Wei, **Wesley W. Qian**, Kelsie Little, Matthew Andres, Britney B. Nguyen, Theresa Moloy, Jacob Yasonik, Jane K. Parker, Richard C. Gerkin, Joel D. Mainland, Alexander B. Wiltchko

- **3D equivariant diffusion for target-aware molecule generation and affinity prediction**

ICLR (2023)

Jiaqi Guan*, **Wesley W. Qian***, Xingang Peng, Yufeng Su, Jian Peng, Jianzhu Ma

- **Metabolic activity organizes olfactory representations**

eLife (2023)

Wesley W. Qian, Jennifer N. Wei, Benjamin Sanchez-Lengeling, Brian K. Lee, Yunan Luo, Marnix Vlot, Koen Dechering, Jian Peng, Richard C. Gerkin, Alexander B. Wiltchko

- **A central chaperone-like role for 14-3-3 proteins in human cells**

Molecular Cell (2023)

Dmitri Segal, Stefan Maier, Giovanni J Mastromarco, **Wesley W. Qian**, Syed Nabeel-Shah, Hyunmin Lee, Gaelen Moore, Jessica Lacoste, Brett Larsen, Zhen-Yuan Lin, Abeeshan Selvakumaran, Karen Liu, Craig Smibert, Zhaolei Zhang, Jack Greenblatt, Jian Peng, Hyun O Lee, Anne-Claude Gingras, Mikko Taipale

- **Energy-inspired molecular conformation optimization**

ICLR (2022)

Jiaqi Guan*, **Wesley W. Qian***, Qiang Liu, Wei-Ying Ma, Jianzhu Ma, Jian Peng

- **ECNet is an evolutionary context-integrated deep learning framework for protein engineering**
Nature Communications (2021)
Yunan Luo, Guangde Jiang, Tianhao Yu, Yang Liu, Lam Vo, Hantian Ding, Yufeng Su, **Wesley W. Qian**, Huimin Zhao, Jian Peng
- **Integrating deep neural networks and symbolic inference for organic reactivity prediction**
ACS National Meeting (2021)
Wesley W. Qian*, Nathan T. Russell*, Claire L. W. Simons, Yunan Luo, Martin D. Burke, Jian Peng
- **Comprehensive interactome profiling of the human Hsp70 network highlights functional differentiation of J domains**
Molecular Cell (2021)
Benjamin L. Piette, Nader Alerasool, Zhen-Yuan Lin, Jessica Lacoste, Mandy Hiu Yi Lam, **Wesley W. Qian**, Stephanie Tran, Brett Larsen, Eric Campos, Jian Peng, Anne-Claude Gingras, Mikko Taipale
- **Evaluating attribution for graph neural networks**
NeurIPS (2020)
Benjamin Sanchez-Lengeling, Jennifer Wei, Brian Lee, Emily Reif, Peter Wang, **Wesley W. Qian**, Kevin McCloskey, Lucy Colwell, Alexander B. Wiltschko
- **Batch equalization with a generative adversarial network**
Bioinformatics (2020)
Wesley W. Qian, Cassandra Xia, Subhashini Venugopalan, Arunachalam Narayanaswamy, Michelle Dimon, George W. Ashdown, Jake Baum, Jian Peng, D Michael Ando
- **Evolutionary context-integrated deep sequence modeling for protein engineering**
RECOMB (2020)
Yunan Luo, Lam Vo, Hantian Ding, Yufeng Su, Yang Liu, **Wesley W. Qian**, Huimin Zhao, Jian Peng

Services

- **Reviewer** for Nature Machine Intelligence (2025), ICML (2024), ICLR (2024), NeurIPS (2023), LoG (2022 & 2023), RECOMB (2021), ISMB (2019 & 2020).
- **Program Committee** for ICML: ML Interpretability for Scientific Discovery Workshop 2020.